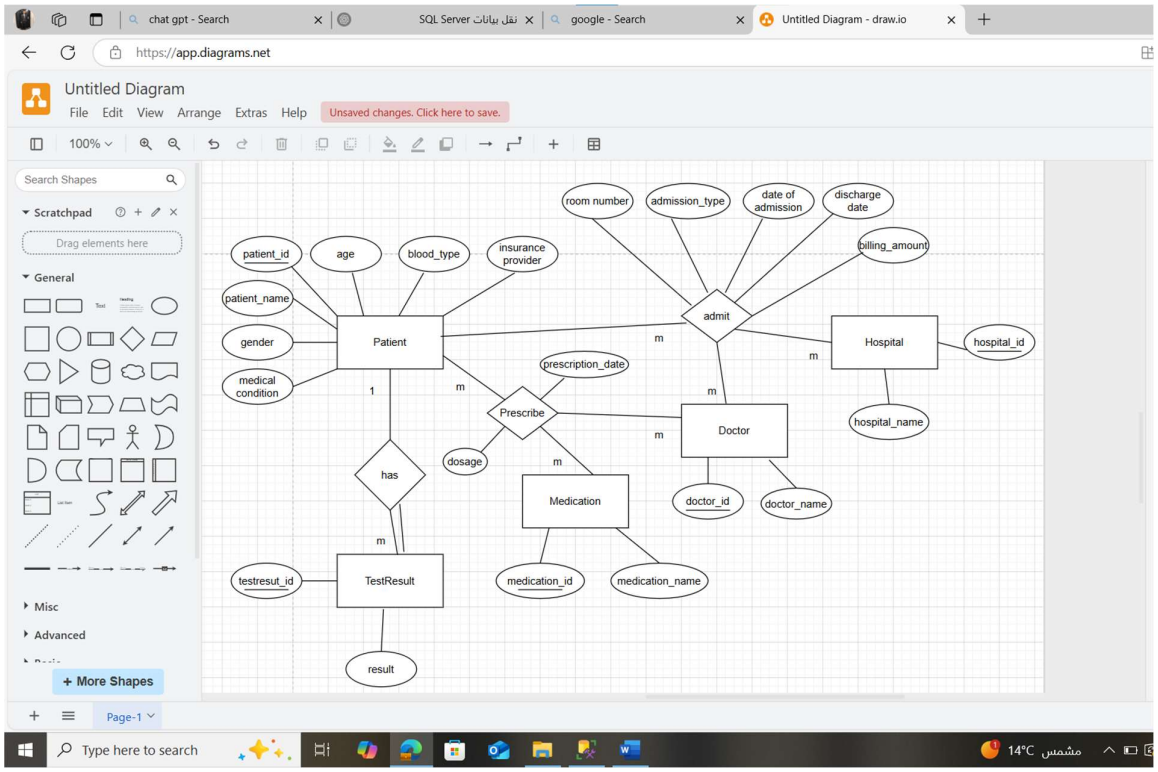
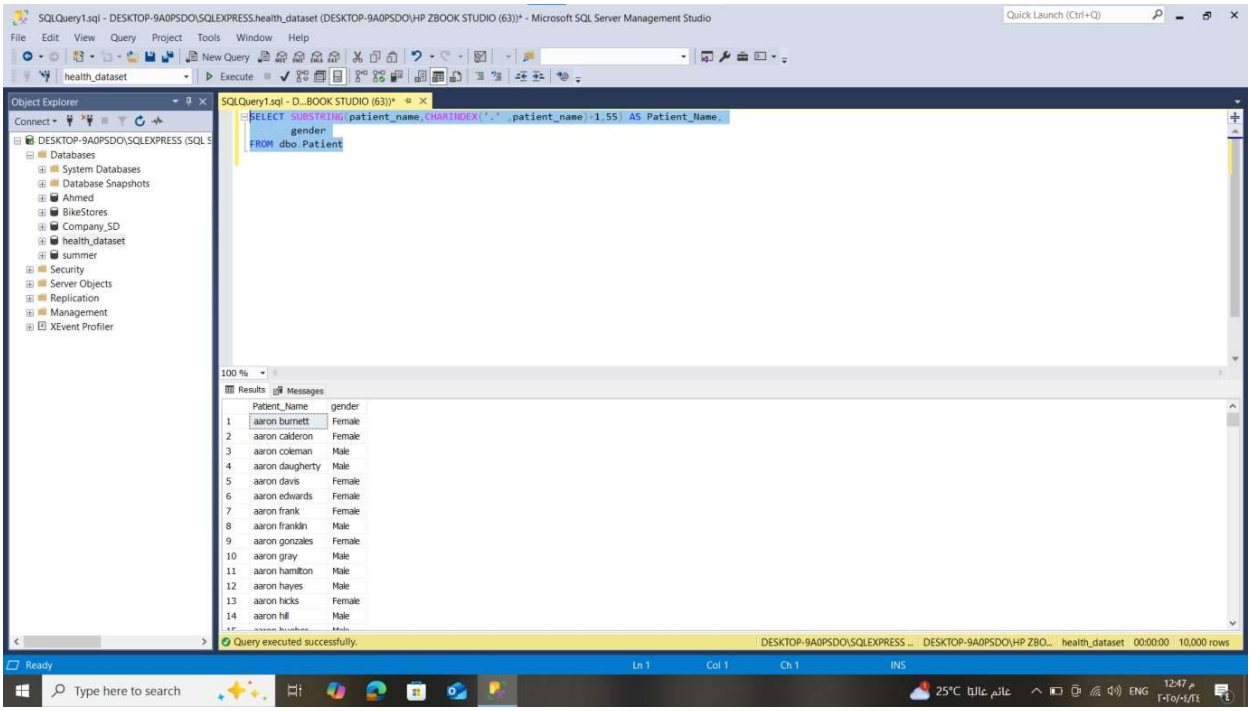
SQL PROJECT I

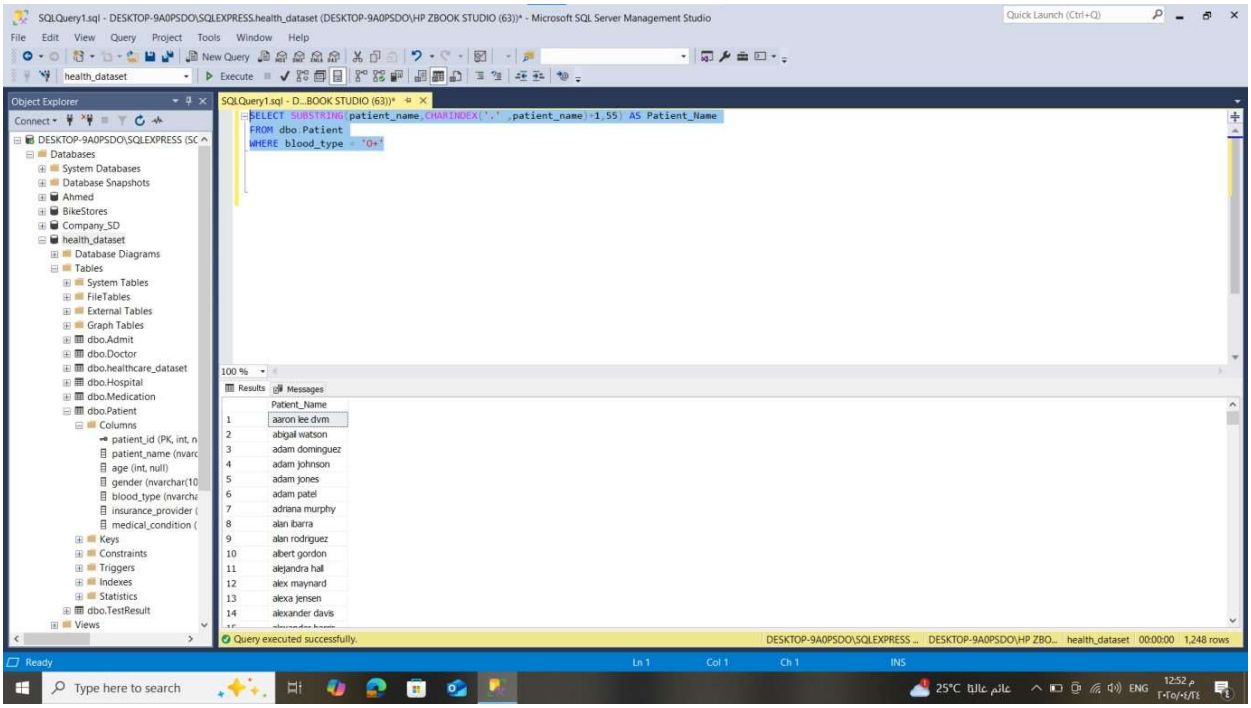


1. Retrieve the names and genders of all patients.



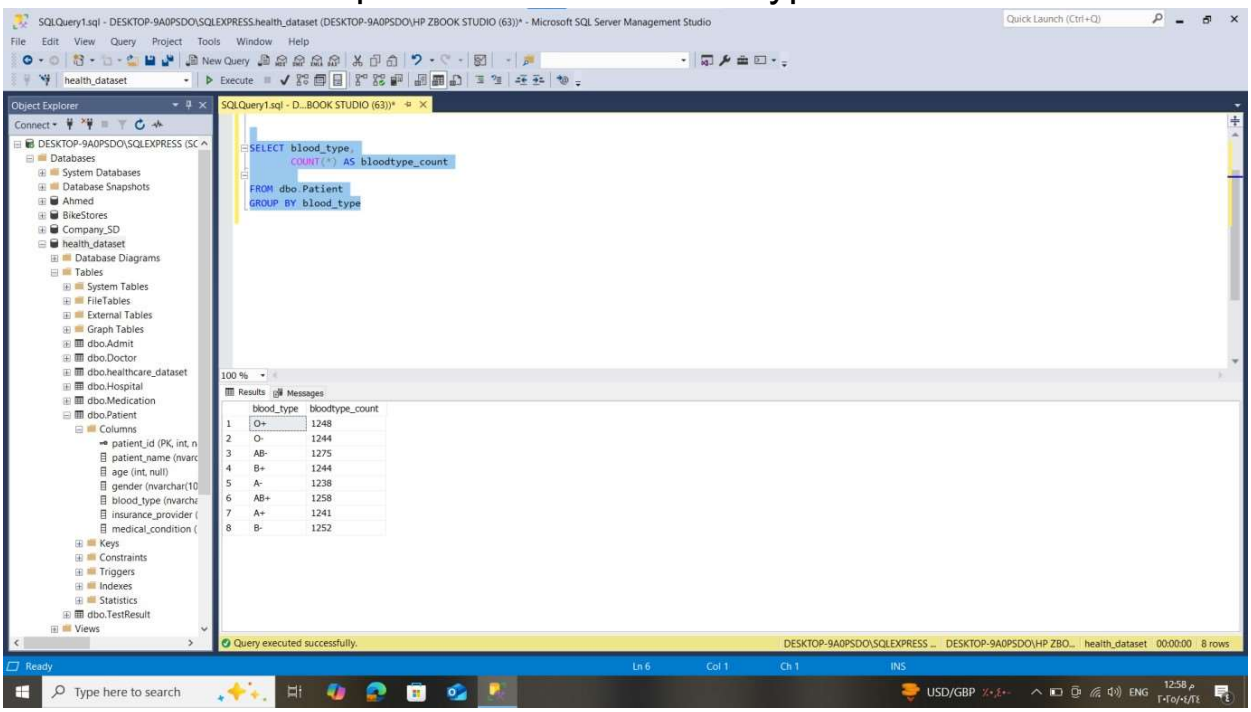
2. Display all patients with blood type O+.

3.

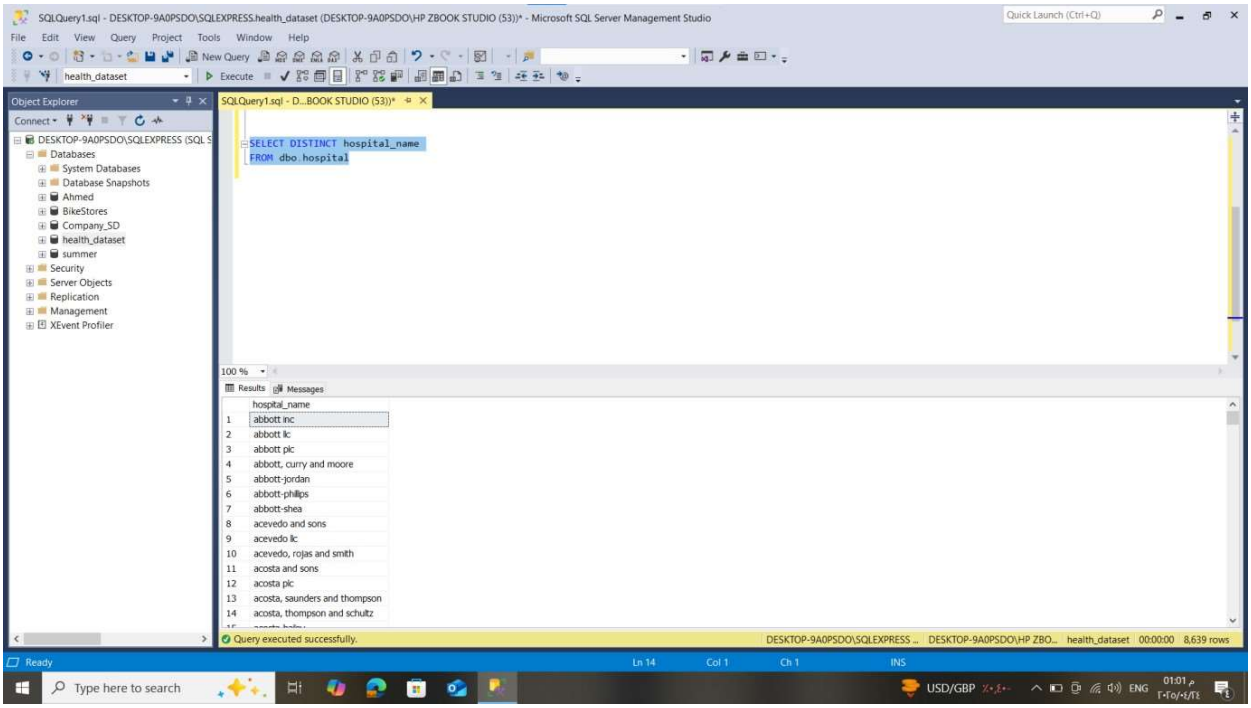


Count the number of patients in each blood type.

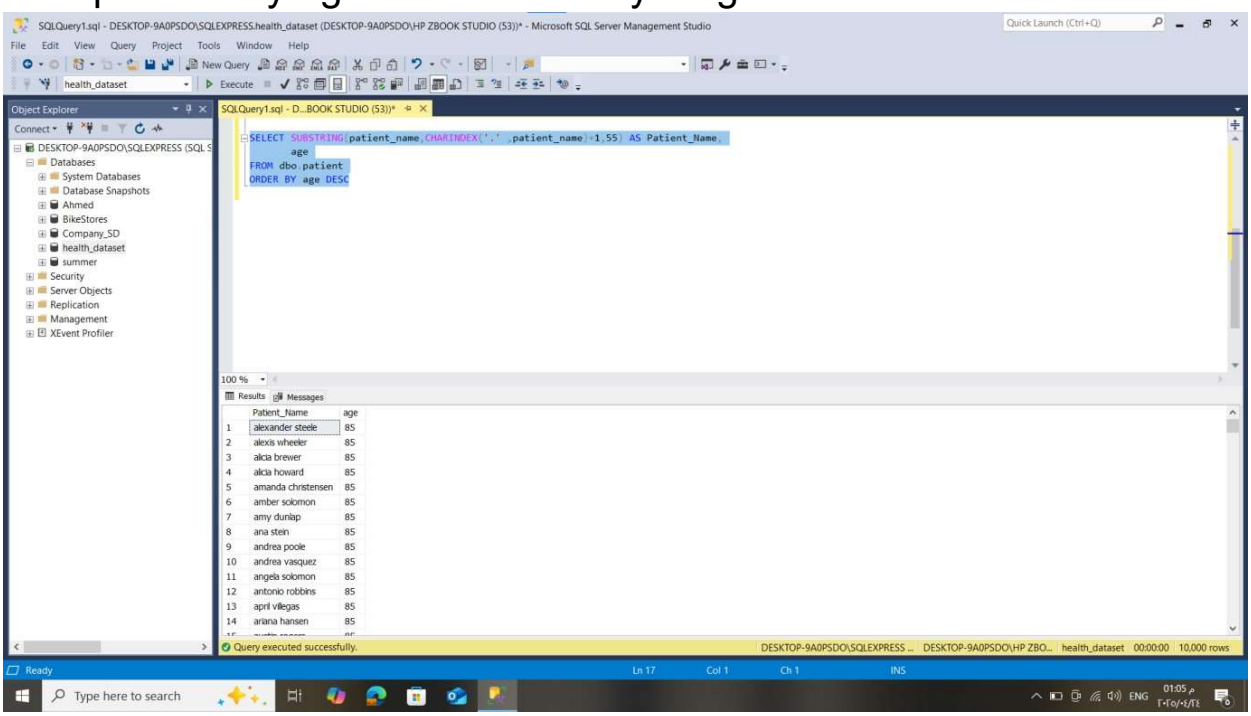
4.



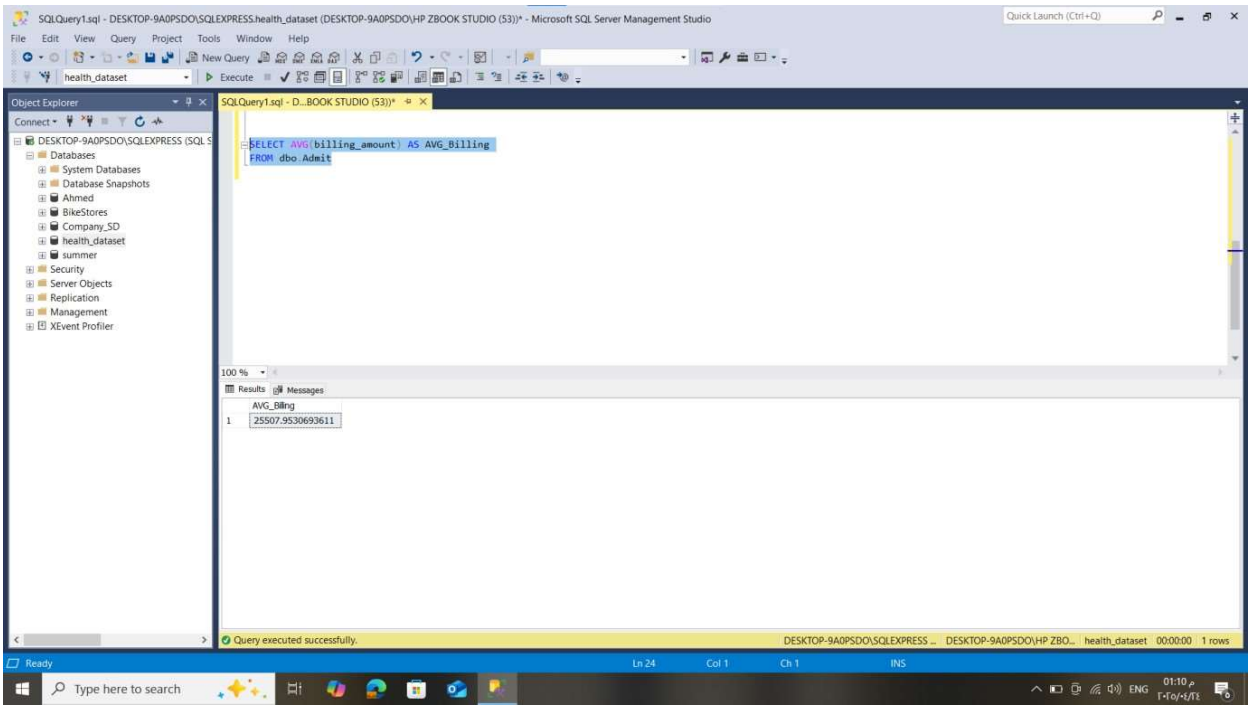
List all distinct hospital names.

5. 

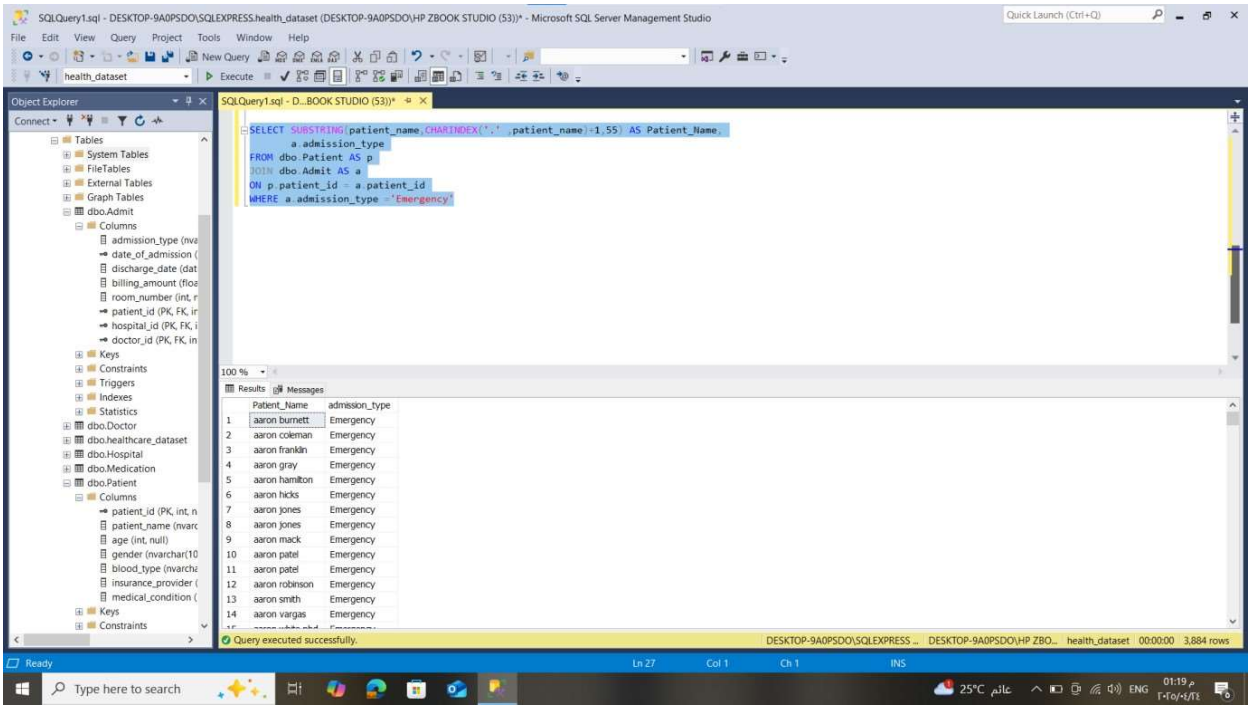
Order patients by age from oldest to youngest.

6. 

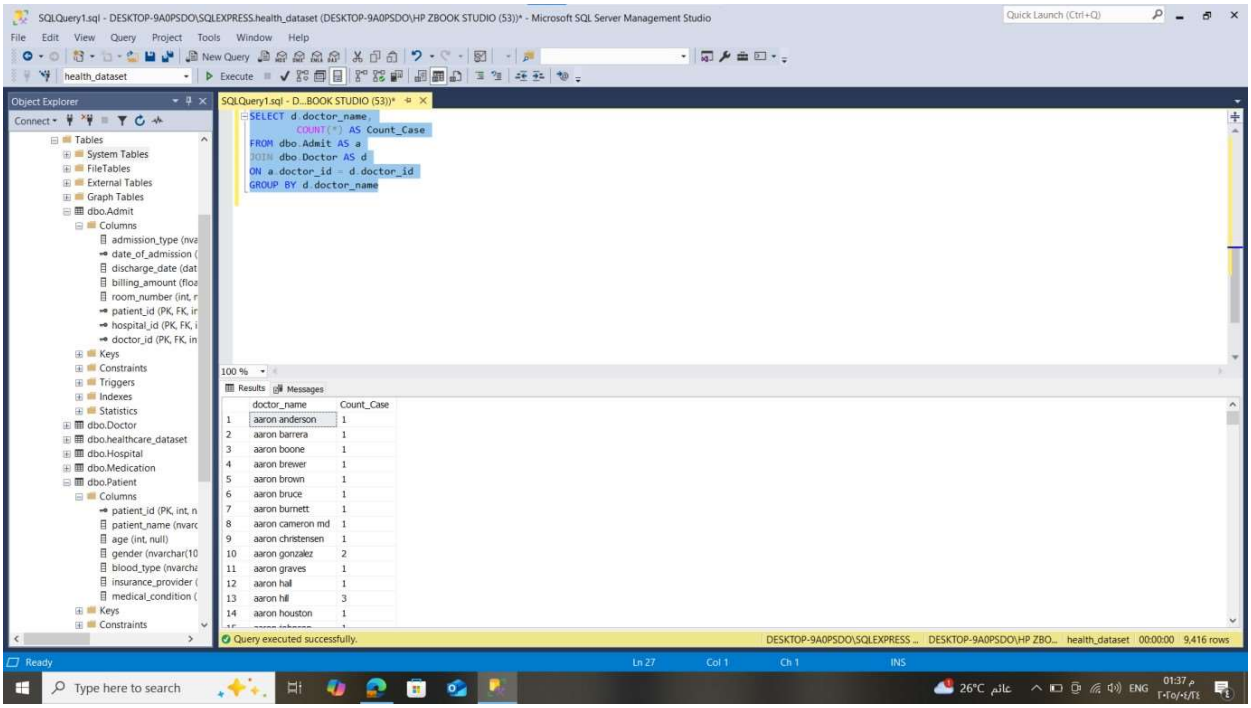
Calculate the average billing amount for all admissions.

7. 

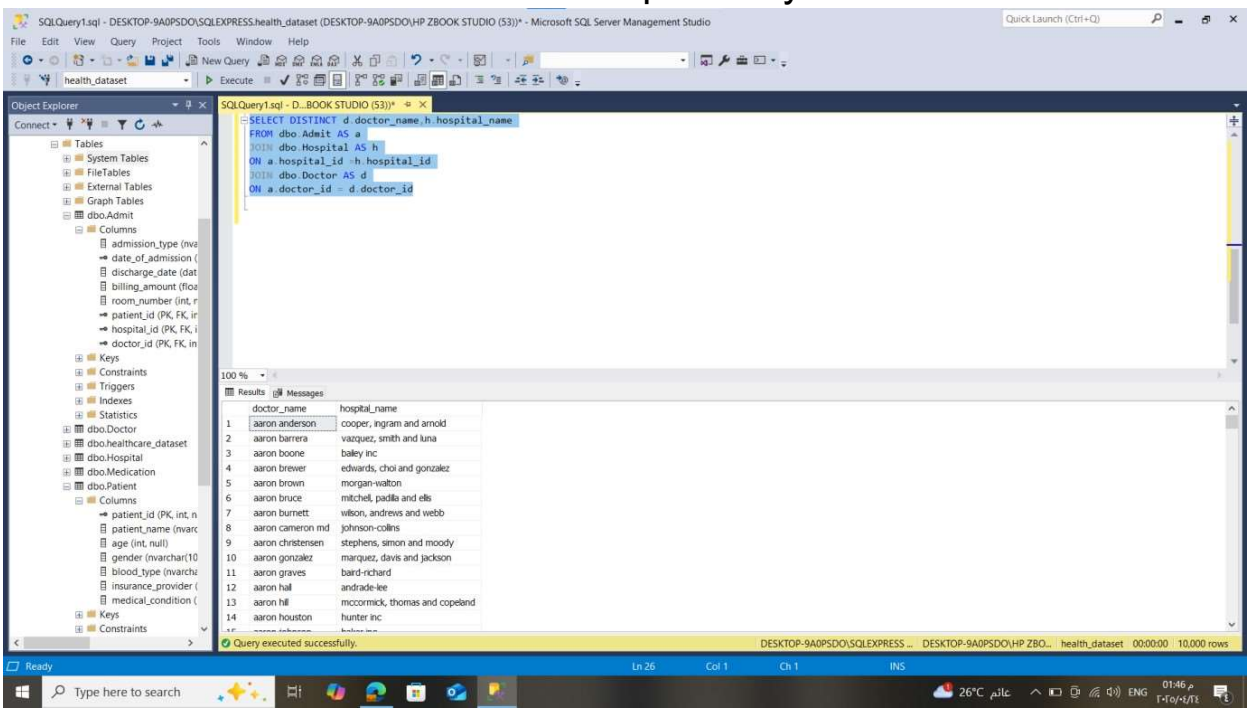
Display patients admitted through 'Emergency'.

8. 

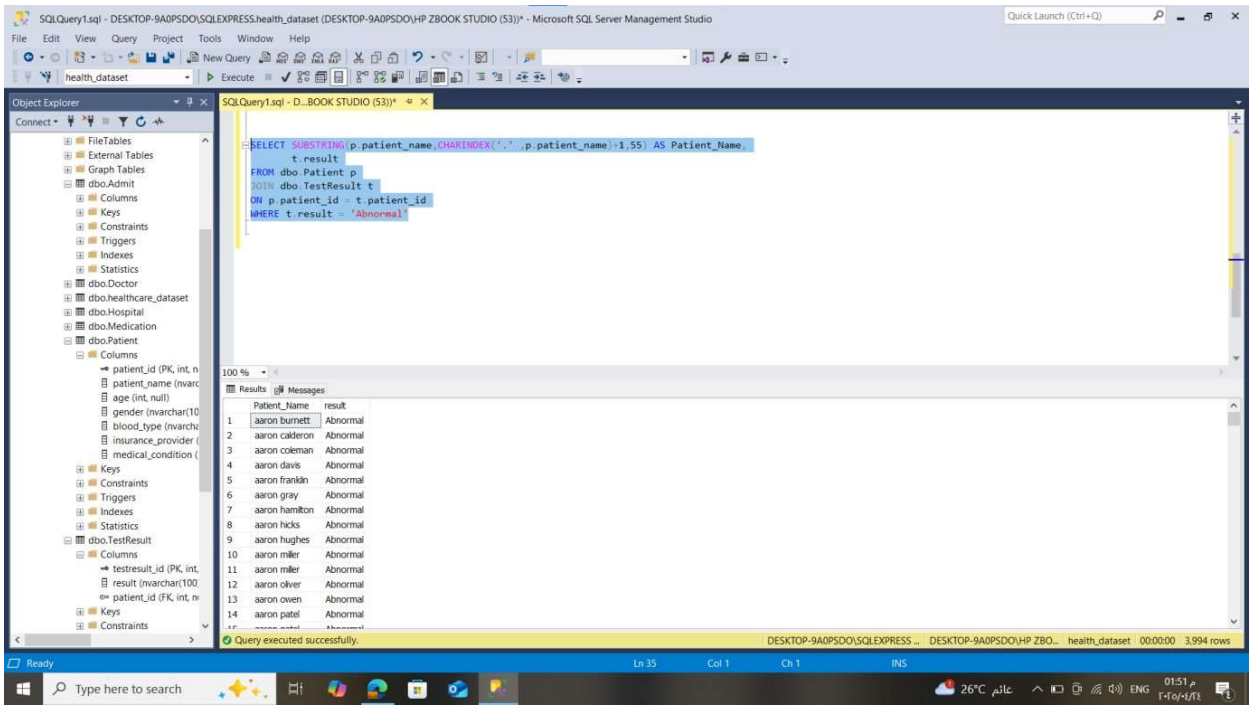
Count the number of cases handled by each doctor.

9. 

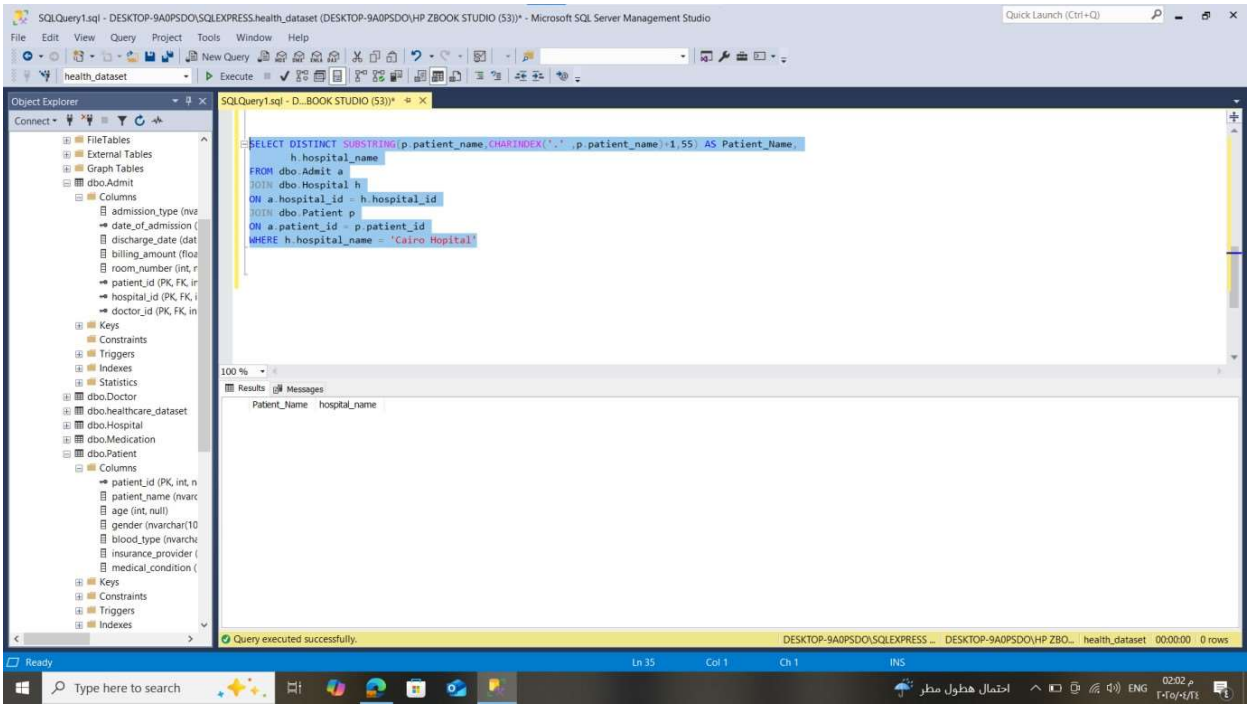
Show the doctor name and the hospital they work in.

10. 

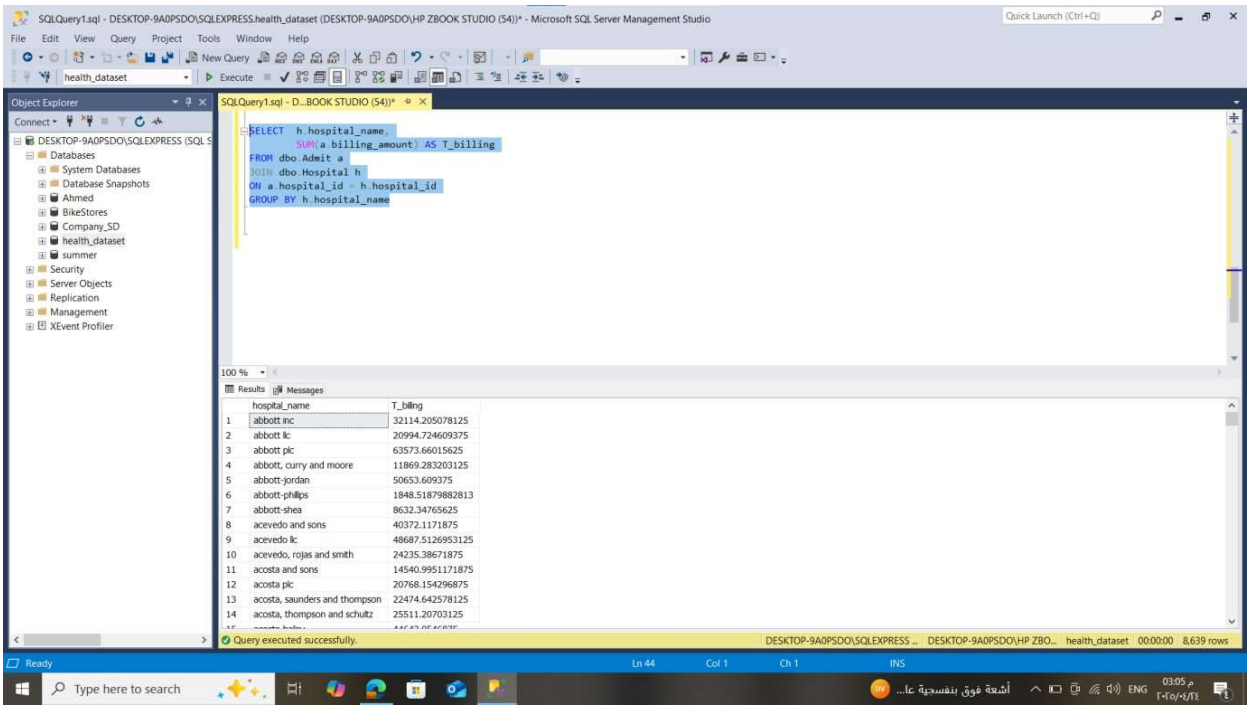
List patients whose test results are 'Abnormal'.



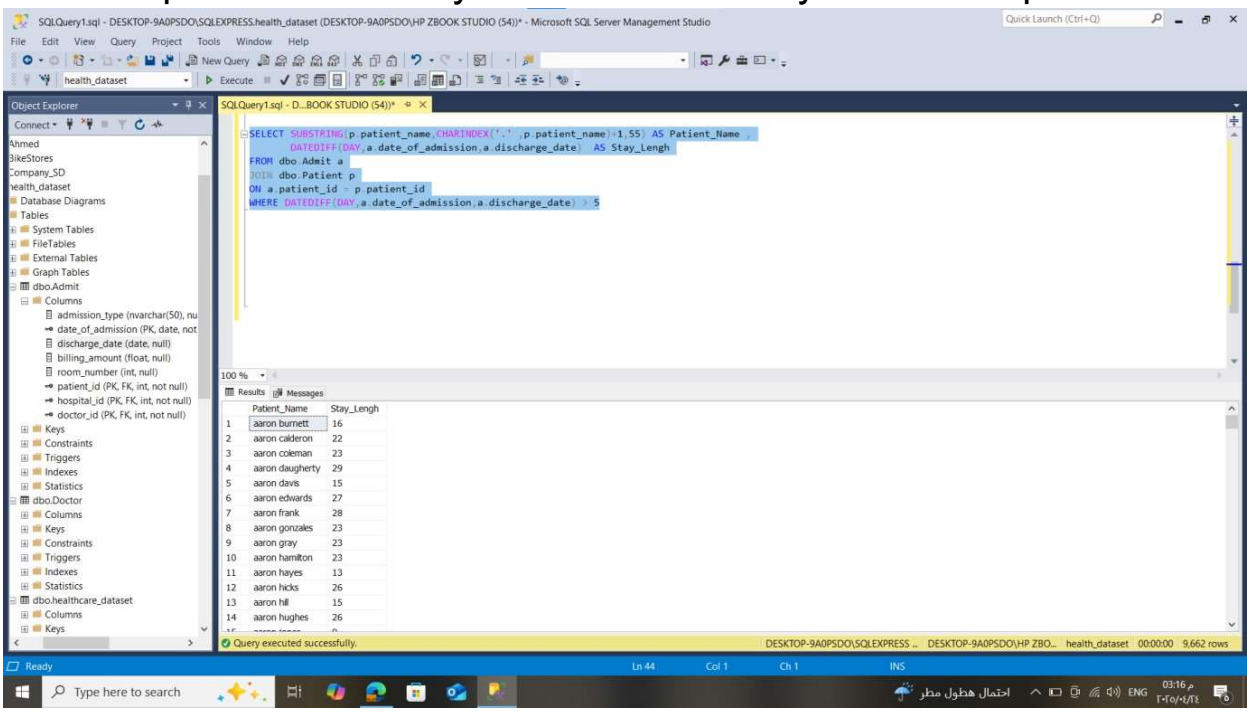
11. Display all patients treated in a hospital named 'Cairo Hospital' using JOIN.



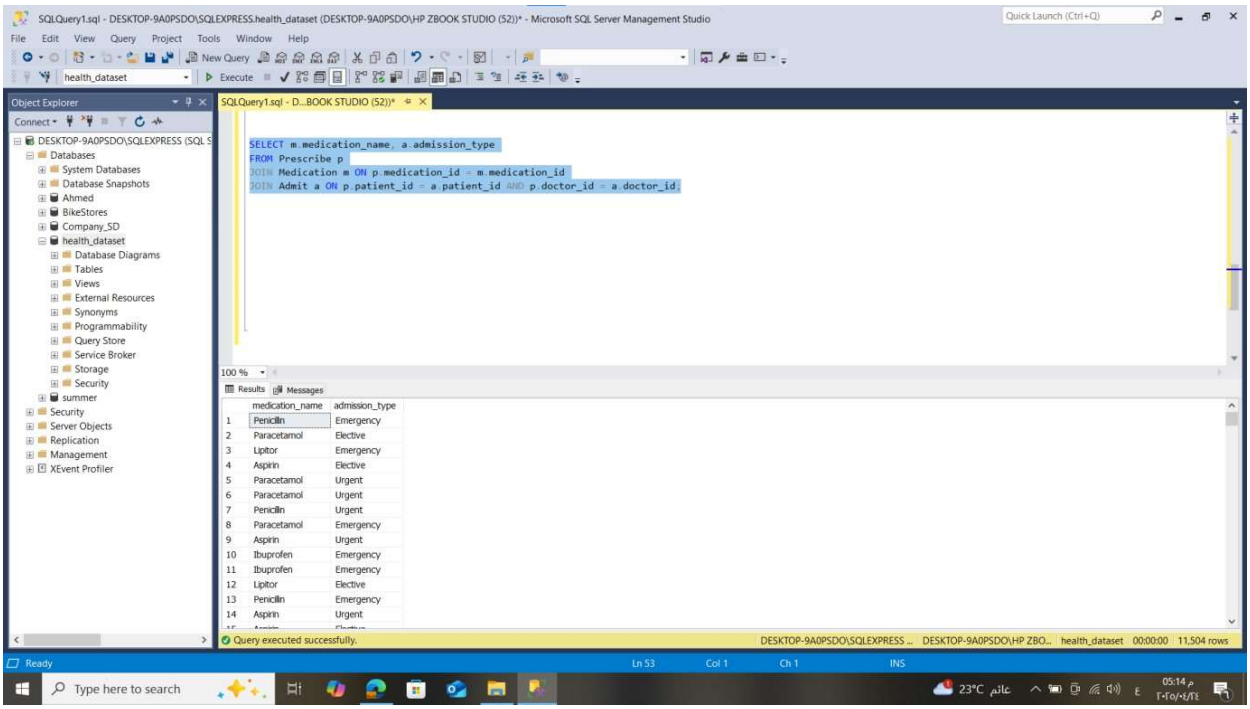
12. Calculate the total billing amount per hospital.



13. List the patients who stayed more than 5 days in the hospital.



14. Show all medications used per admission using JOIN.



15. Count the number of patients per medical condition.

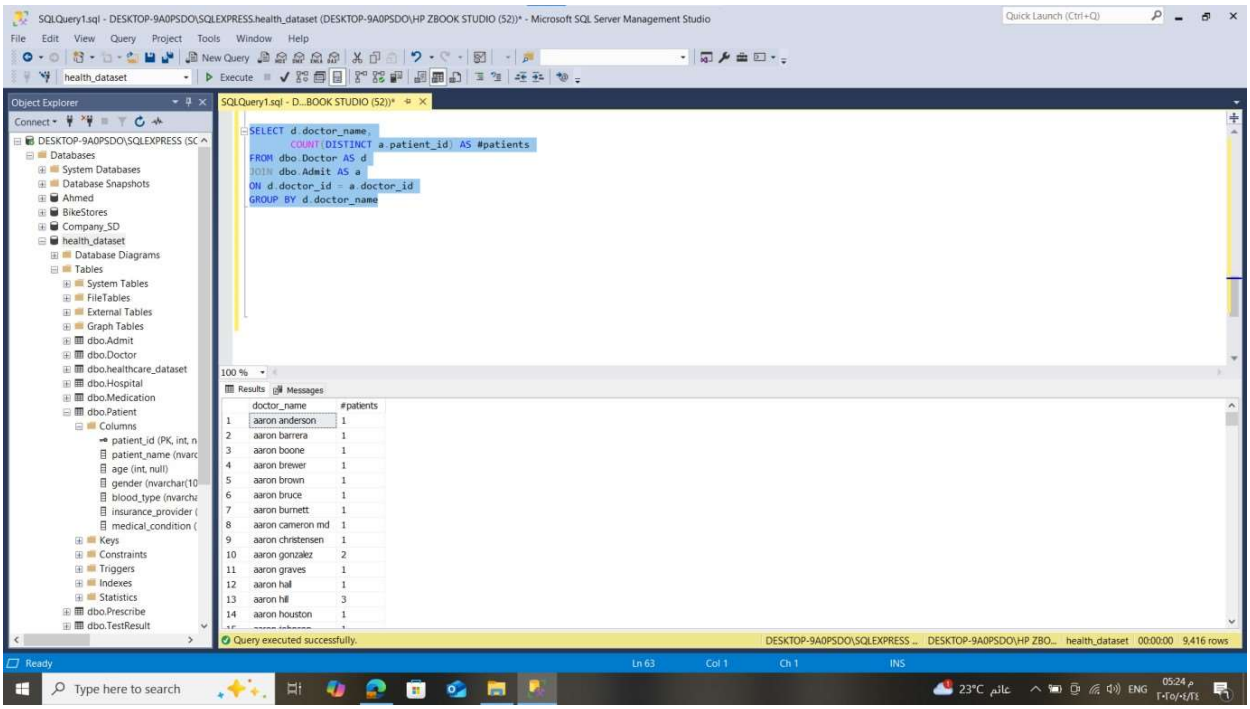
```
SELECT medical_condition,
       COUNT(*) AS #Patient
FROM dbo.Patient
GROUP BY medical_condition
```

100 %

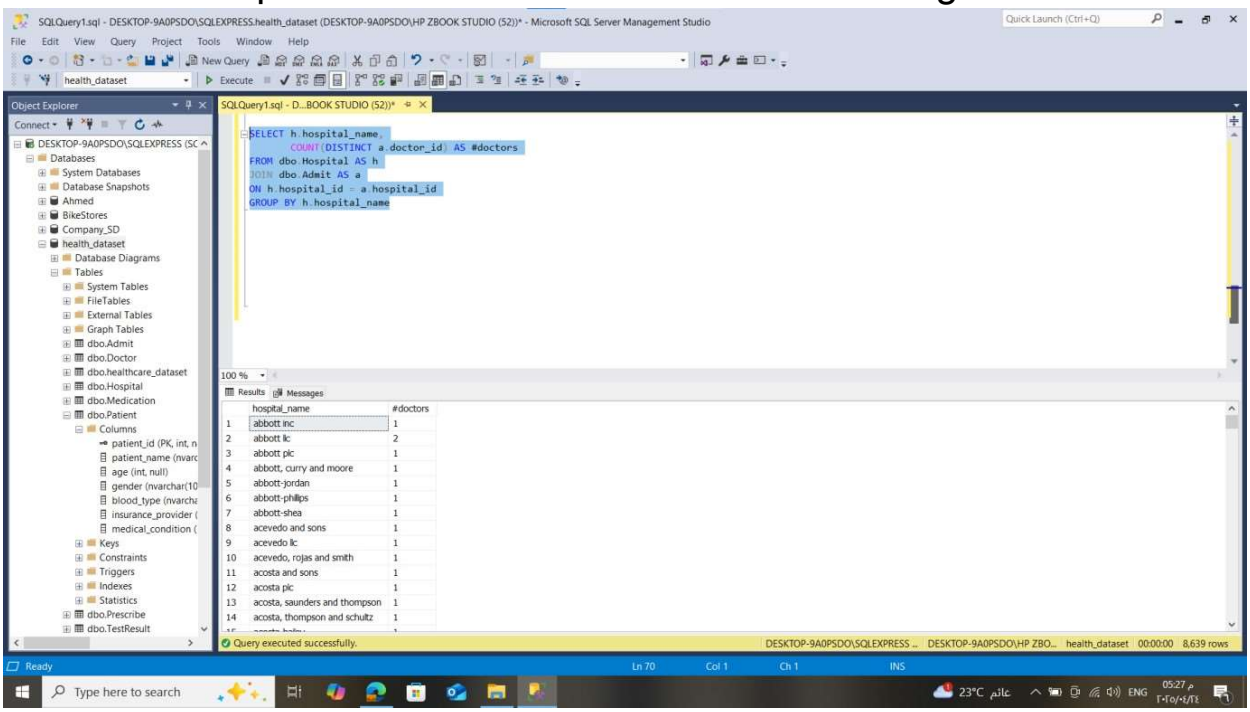
Results | Messages

	medical_condition	#Patient
1	Diabetes	4869
2	Cancer	5109
3	Arthritis	4950
4	Asthma	5124
5	Hypertension	5064
6	Obesity	4884

16. Show each doctor and how many patients they treated.



17. List each hospital and the number of doctors working there.



18. Display patients discharged before '2024-01-01'.

```sql
SELECT SUBSTRING(p.patient_name,CHARINDEX('.' ,p.patient_name)+1,55) AS Patient_Name ,
       a.discharge_date
FROM dbo.Admit a
JOIN dbo.Patient p
ON a.patient_id = p.patient_id
WHERE YEAR(a.discharge_date) < 2024
```

100 %

Results | Messages

	Patient_Name	discharge_date
1	aaron burnett	2021-06-19
2	aaron calderon	2023-06-03
3	aaron coleman	2019-10-10
4	aaron daugherty	2022-05-08
5	aaron davis	2019-07-13
6	aaron edwards	2021-01-05
7	aaron frank	2021-05-05
8	aaron franklin	2020-10-10
9	aaron gonzales	2019-10-03
10	aaron gray	2021-06-24
11	aaron hamilton	2022-09-13
12	aaron hayes	2019-02-11
13	aaron hicks	2023-05-26
14	aaron hill	2022-01-23

Query executed successfully. DESKTOP-9A0PS

19. Show patients with billing amounts above the average (use subquery).

```sql
SELECT SUBSTRING(p.patient_name,CHARINDEX('.' ,p.patient_name)+1,55) AS Patient_Name ,
       a.billing_amount,
       AVG(billing_amount) AS AVG_Billing
FROM dbo.Admit a
JOIN dbo.Patient p
ON a.patient_id = p.patient_id
GROUP BY SUBSTRING(p.patient_name,CHARINDEX('.' ,p.patient_name)+1,55),
       a.billing_amount
HAVING A.billing_amount > (SELECT AVG(billing_amount) AS AVG_Billing
                           FROM dbo.Admit )
```

100 %

Results | Messages

	Patient_Name	billing_amount	AVG_Billing
1	heather johnston	25511.20703125	25511.20703125
2	jason bailey	25521.44921875	25521.44921875
3	robert waters	25524.9921875	25524.9921875
4	tracey carroll	25529.9140625	25529.9140625
5	melissa atkins	25543.611328125	25543.611328125
6	timothy henderson	25553.109375	25553.109375
7	sandra cunningham	25558.314453125	25558.314453125
8	tim chandler	25595.8515625	25595.8515625
9	melvin miller	25598.763671875	25598.763671875
10	patricia morris	25600.23046875	25600.23046875
11	alexander sellers	25610.642578125	25610.642578125
12	mathew edwards	25622.162109375	25622.162109375
13	jason mcguire	25629.265625	25629.265625
14	maria clark	25632.673828125	25632.673828125

Query executed successfully. DESKT

Ln 84 Col 1 Ch

20. List patients who were given the medication 'Aspirin'.

```sql
SELECT DISTINCT SUBSTRING(p.patient_name,CHARINDEX('.' ,p.patient_name)+1,55) AS Patient_Name ,m.medication_name
FROM dbo.Prescribe pe
JOIN dbo.Medication m
ON pe.medication_id = m.medication_id
JOIN dbo.Patient p
ON pe.patient_id = p.patient_id
WHERE m.medication_name = 'Aspirin'
```

100 %

Results | Messages

	Patient_Name	medication_name
1		Aspirin
2	benjamin jimenez	Aspirin
3	bobby johnson	Aspirin
4	brian cordova	Aspirin
5	christina hodges md	Aspirin
6	david gibbs jr.	Aspirin
7	deanna rodriguez	Aspirin
8	derrick robinson	Aspirin
9	donna bishop	Aspirin
10	emily baxter	Aspirin
11	gregory pitts	Aspirin
12	hayden rubio	Aspirin
13	james coleman dds	Aspirin
14	janet ramos	Aspirin

Query executed successfully. DESKTOP-9A0PSDO\SQLEXPRESS ... DESKTOP-9A0PSDO\HP ZBO...

Ln 96 Col 1 Ch 1 INS

21. Show each patient's name, their doctor's name, and hospital name. If the medical condition is NULL, show 'No Diagnosis' (use COALESCE with JOINs across patients, doctors, and hospitals).

```sql
SELECT d.doctor_name,
       h.hospital_name,
       SUBSTRING(p.patient_name,CHARINDEX('.' ,p.patient_name)+1,55) AS Patient_Name ,
       COALESCE(P.medical_condition,'No Diagnosis') AS Medical_Condition
FROM dbo.Admit a
JOIN dbo.Doctor d
ON a.doctor_id = d.doctor_id
JOIN dbo.Hospital h
ON a.hospital_id = h.hospital_id
JOIN dbo.Patient p
ON a.patient_id = p.patient_id
```

100 %

Results | Messages

	doctor_name	hospital_name	Patient_Name	Medical_Condition
1	michael banks	brooks, wright and olson	matthew thompson	Cancer
2	kevin collins	bentley-thomas	sherry ho	Asthma
3	kristen hancock	brown-green	christopher cervantes	Hypertension
4	maria kelly	brady llc	tim green	Hypertension
5	kathryn sharp	lam, rodriguez and berger	lisa stewart	Hypertension
6	kathryn sharp	lam, rodriguez and berger	lisa stewart	Cancer
7	jason hill	white-hahn	richard romero	Obesity
8	kristie curtis	martinez and sons	wesley rogers	Hypertension
9	julia murray	martinez and sons	terrance knapp	Diabetes
10	john brown	silva, garcia and hicks	brooke anderson	Hypertension
11	mario wright	frye, patton and carter	anne dougherty	Asthma
12	christopher davis	brown, duncan and hughes	melissa lozano	Obesity
13	david bradley	morris-shepherd	amber thomas	Obesity
14	grant johnson	rodriguez, barrett and thompson	thomas webb	Diabetes

22. List patients who were admitted more than once, showing their name and number of admissions (use GROUP BY and HAVING COUNT(*) > 1).

```sql
SELECT  SUBSTRING(p.patient_name,CHARINDEX('.' ,p.patient_name)+1,55) AS Patient_Name , COUNT(*) AS admission_count
FROM Admit a
JOIN Patient p ON a.patient_id = p.patient_id
GROUP BY SUBSTRING(p.patient_name,CHARINDEX('.' ,p.patient_name)+1,55)
HAVING COUNT(*) > 1;
```

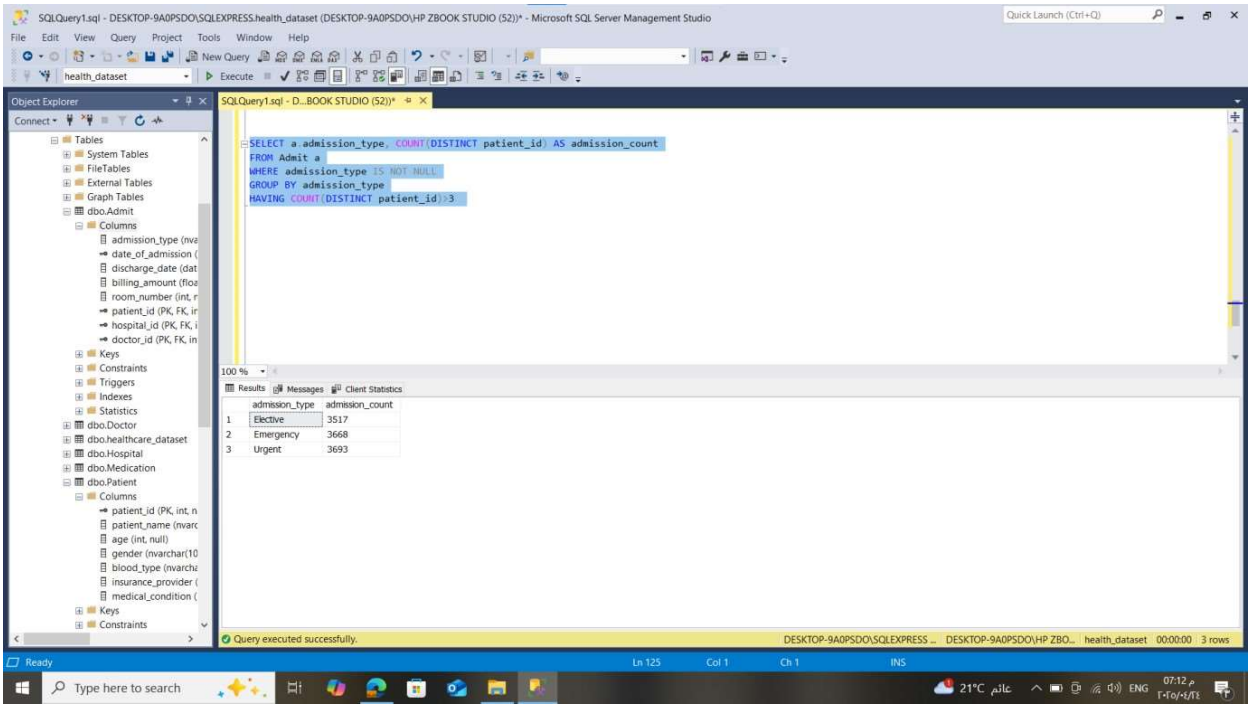
100 %

Results | Messages

	Patient_Name	admission_count
1	john brown	4
2	alicia howard	4
3	adam callahan	4
4	timothy hernandez	4
5	michael lewis	9
6	travis walker	4
7	nicole evans	4
8	daniel alexander	4
9	matthew norris	4
10	amy brown	9
11	jeffrey mcdonald	4
12	william smith	4
13	ryan smith	4
14	beverly johnson	4

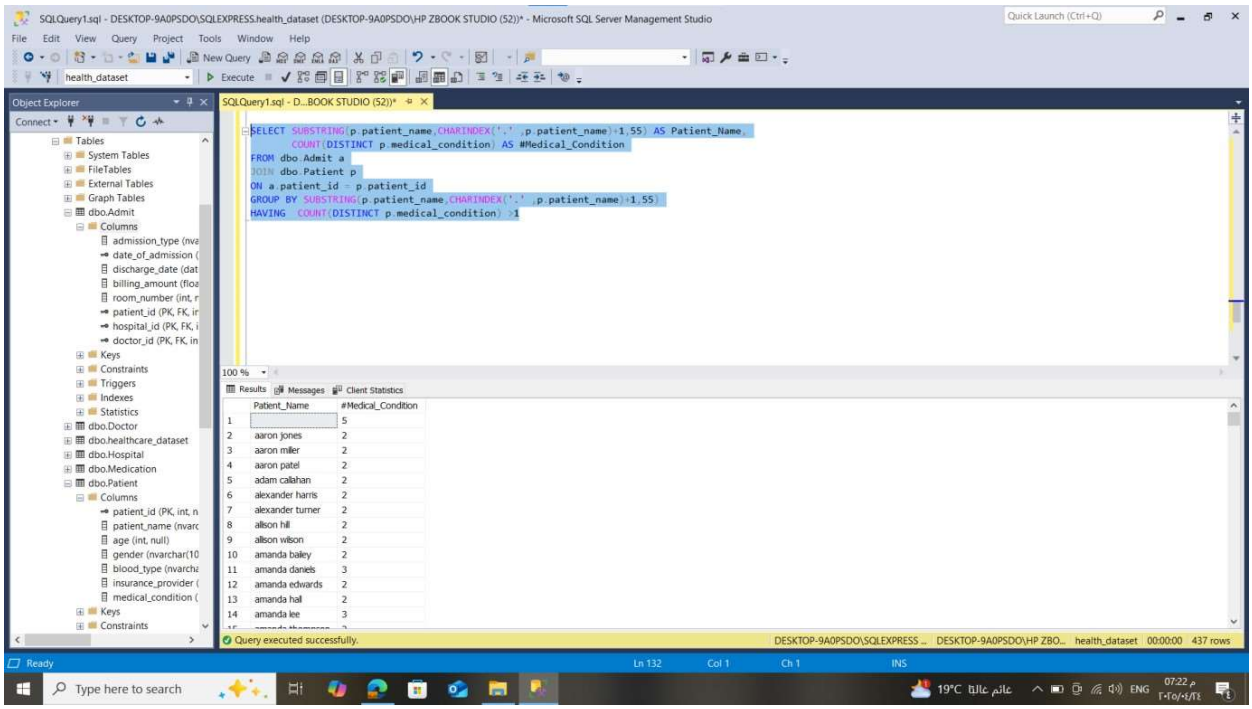
Query executed successfully. DESKTOP-9A0PSDO\SQLEXPRESS ...

23. Count the number of patients per admission_type, only showing those with more than 3 and where admission_type is not NULL.



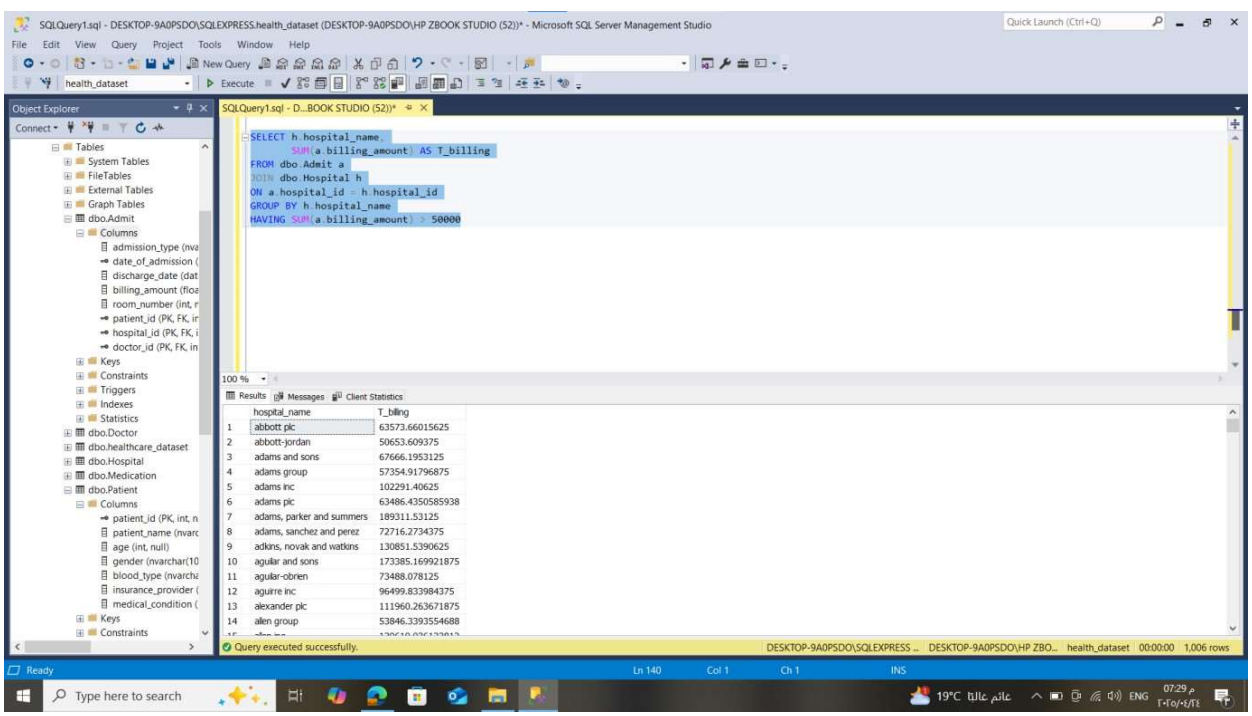
24. List patients with more than one distinct medical condition using

JOINs across patients, admissions, and medical conditions.



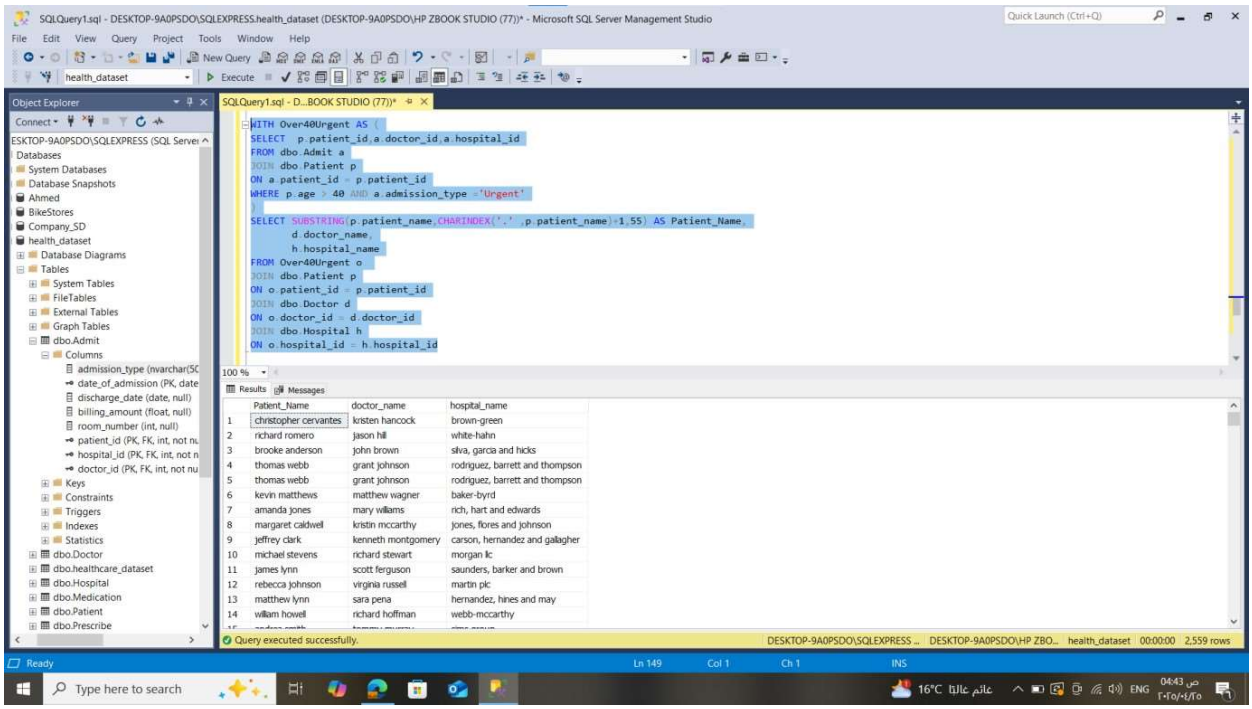
25. Show hospitals where the total billing is more than 50,000 EGP

and display the total billing amount (use GROUP BY + SUM).



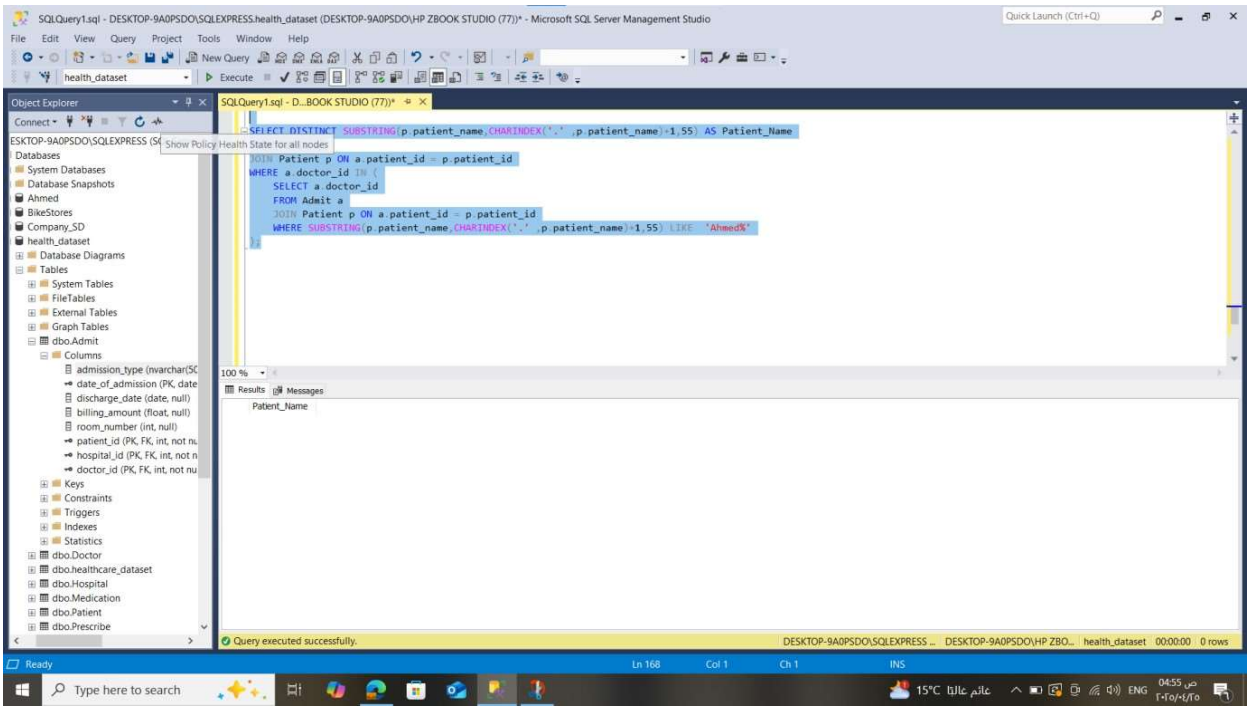
26. Using a CTE, show patients over age 40 who were admitted as

'Urgent', along with their doctor and hospital name.

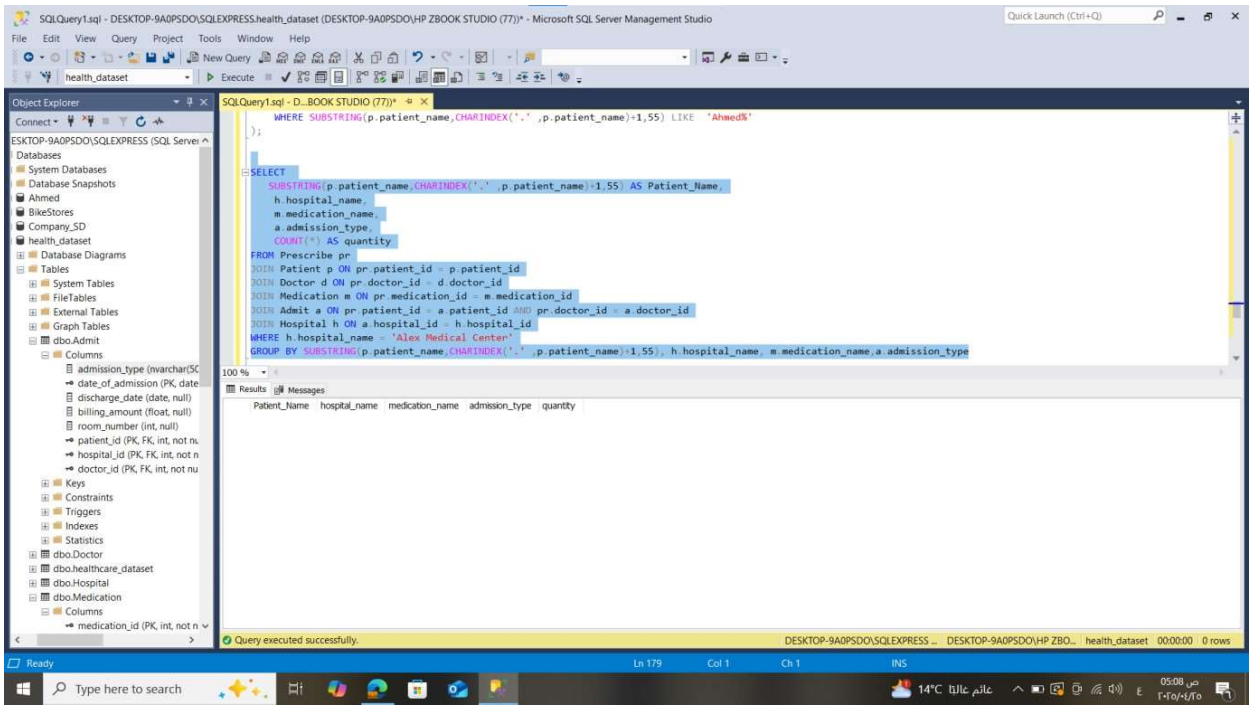


27. Use a subquery to find all patients treated by the same doctor who treated a patient named 'Ahmed'.

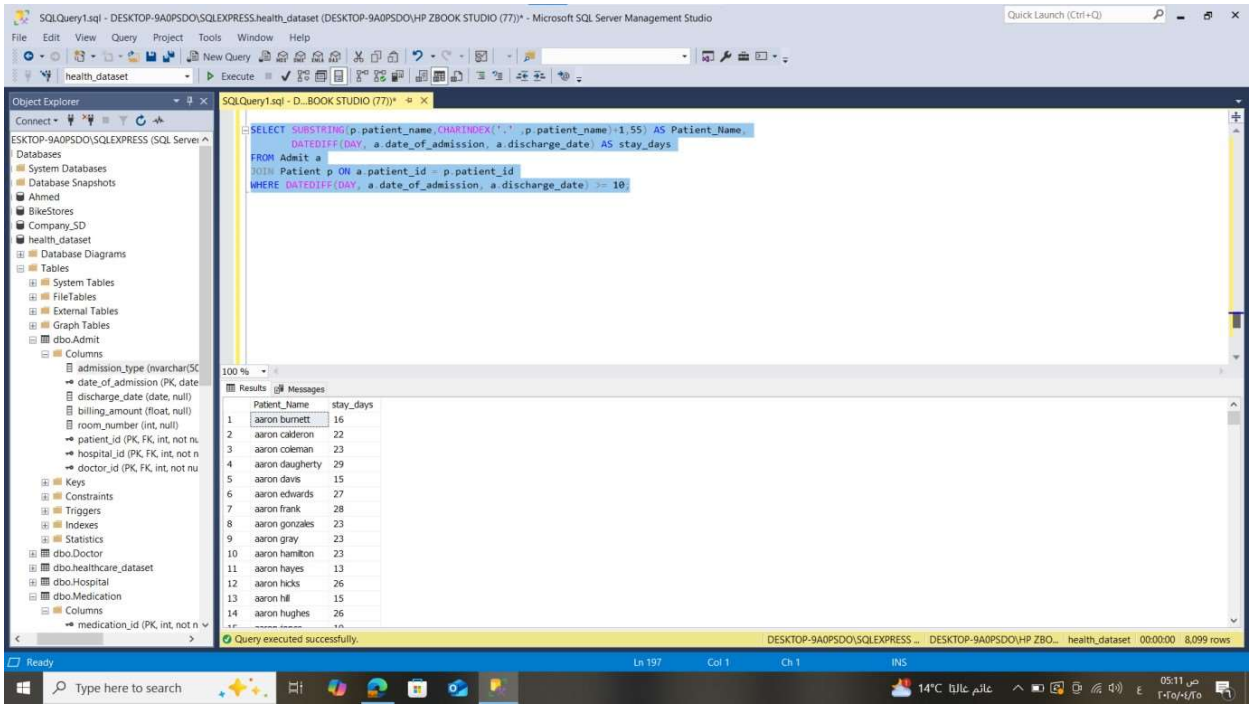
28.



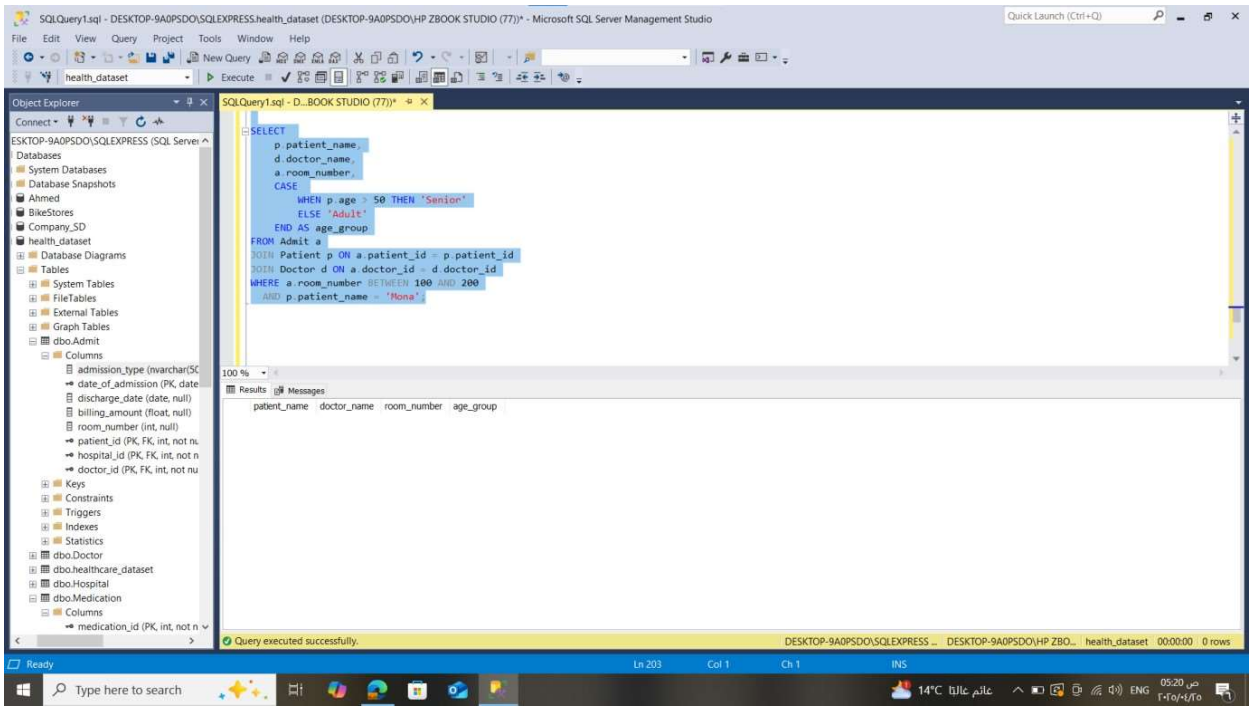
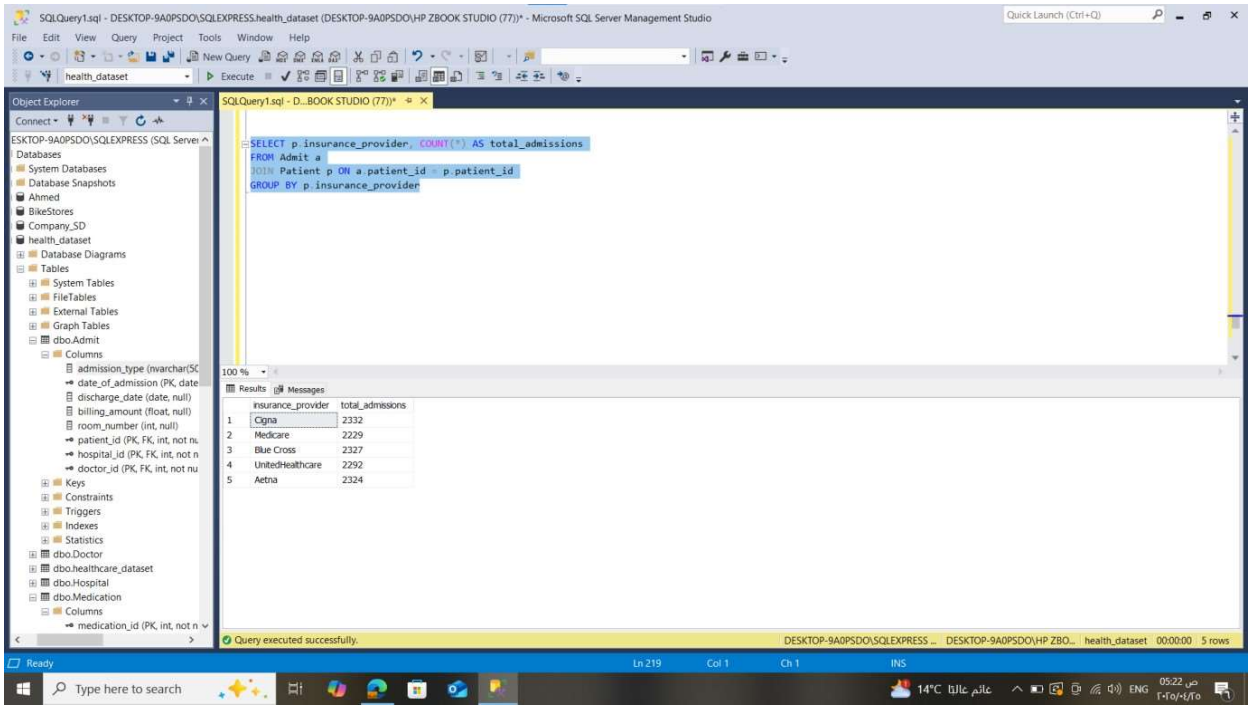
Show all admissions at 'Alex Medical Center' with the medication names and quantities using joins across admissions, hospitals, and medications.



29. List patients who were discharged 10 or more days after admission (use DATEDIFF or date arithmetic).



30. Show patients admitted to rooms between 100 and 200, treated by 'Dr. Mona', and include a column that displays 'Senior' if the patient is older than 50, else 'Adult' (use CASE, JOIN, and filtering).



31.Show the number of admissions per insurance provider.